JFrog Xray RPM Upgrade — Ansible Playbook

Automates upgrading JFrog Xray (3.x → 3.x) on RPM-based systems
(Red Hat, CentOS, Rocky Linux, AlmaLinux).

Files

File	Purpose
`upgrade_xray.yml`	Main Ansible playbook
`inventory.ini`	Inventory template — edit before running

Prerequisites

- Ansible 2.12+ on the control node
- SSH access to Xray host(s) with sudo/become
- Xray host must have internet access to `releases.jfrog.io` (or configure `download_base_url` to point to an internal mirror)
- PostgreSQL storage expanded by **at least 2.5×** before running
- Artifactory 7.x must be running and reachable

Quick Start

1. Edit inventory

```
vi inventory.ini
# Set correct hostnames and SSH user
```

2. Dry run (no changes made)

```
ansible-playbook -i inventory.ini upgrade_xray.yml \
  -e "xray_version=3.107.0" \
  -e "dry_run=true"
```

3. Run the upgrade

```
ansible-playbook -i inventory.ini upgrade_xray.yml \
  -e "xray_version=3.107.0"
```

4. HA rolling upgrade (automatic)

The playbook uses `serial: 1` — it upgrades one node at a time automatically.
No extra flags needed for HA.

Key Variables

Variable	Default	Description
`xray_version`	*(required)*	Target Xray version, e.g. `3.107.0`
`xray_home`	`/opt/jfrog/xray`	Xray installation directory
`xray_user`	`xray`	OS user running Xray
`xray_service`	`xray.service`	systemd service name
`backup_dir`	`/opt/jfrog/backups/xray`	Where pre-upgrade backups are stored
`download_base_url`	JFrog releases CDN	Override for internal mirrors
`artifactory_url`	*(optional)*	Used for connectivity pre-check
`dry_run`	`false`	Set `true` to preview without changes

What the Playbook Does

1. **Pre-flight checks** — version, disk space (≥5 GB), Artifactory connectivity

2. **RabbitMQ feature flags** — enabled before upgrade (required for 3.70+)
3. **Backup** — `system.yaml`, `xray.default`, installed RPM list
4. **Download** — RPM installer script from JFrog releases
5. **Stop service** — graceful shutdown, aborts if service won't stop
6. **Erlang fix** — removes incompatible Erlang package (known issue on certain versions)
7. **Install** — runs RPM installer with `--upgrade` flag
8. **Start & health check** — polls `/api/v1/system/ping` for up to 5 minutes
9. **Validation** — confirms version string and scans logs for errors
10. **Cleanup** — removes downloaded installer

Important Notes

- **Upgrading from 2.x?** You must first upgrade to 3.87.x, then run this playbook for the final hop to the latest version.
- **RabbitMQ 4.x migration**: If moving to Xray ≥ 3.88, also update `JF_PRODUCT_RABBITMQ_HOME` in `/etc/opt/jfrog/xray/xray.default` and consider enabling Quorum Queues in `system.yaml`.
- **PostgreSQL `pg_trgm`**: If using an external PostgreSQL and upgrading to Xray ≥ 3.70, ensure the `pg_trgm` extension is installed manually before running.
- **Backups**: Always verify your system AND database backups are current before running this playbook in production.

Troubleshooting

Symptom	Fix
Service won't start after upgrade	Check `{{ xray_home }}/var/log/xray-service.log`
Erlang conflict errors	Playbook auto-removes; re-run if missed
Health check times out	Increase `retries` in the playbook; check DB connectivity

RabbitMQ won’t start	Manually run `rabbitmqctl enable_feature_flag all` as `xray` user